12 September 2023 19:13

bdd sequences:

bdd above : $\{a_n\} = \{a_n\}_{n=1}^{\infty}$ a <u>fixed</u> real no. K s.t. $a_n \leq K \quad \forall n \in \mathbb{N}$

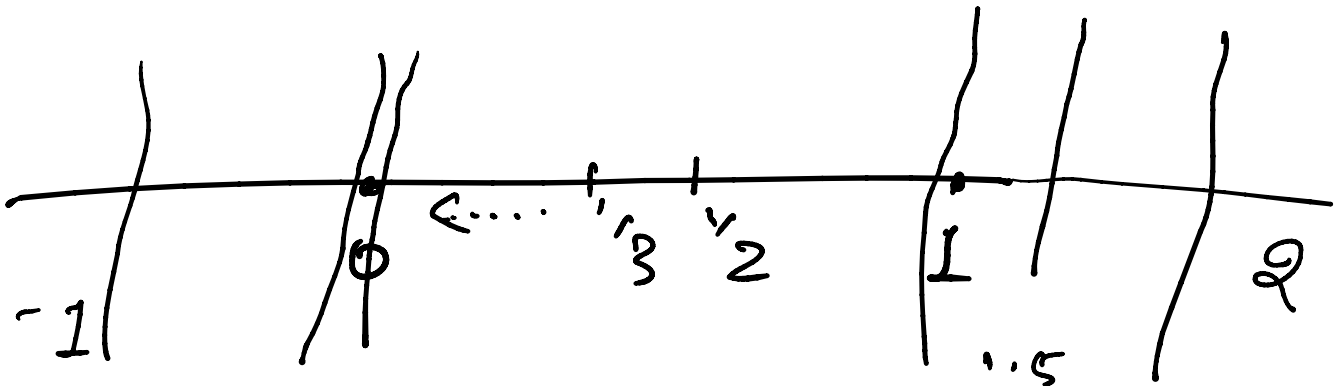
$\{(-1)^n\} = \{-1, 1, -1, 1 \ldots\} \leq 2$

bdd below : $\{a_n\}$ a fixed real no. m s.t. $a_n \geq K \quad \forall n \in \mathbb{N}$

bdd seq : m & K $\quad m \leq a_n \leq K$

$\left\{\frac{1}{n}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3} \ldots\right\}$

$\{x\}$ 1 2 3 5



$(x-1)^2 \geq 0$

$x^2 - 2x + 1 \geq 0$

$2x \leq x^2 + 1$

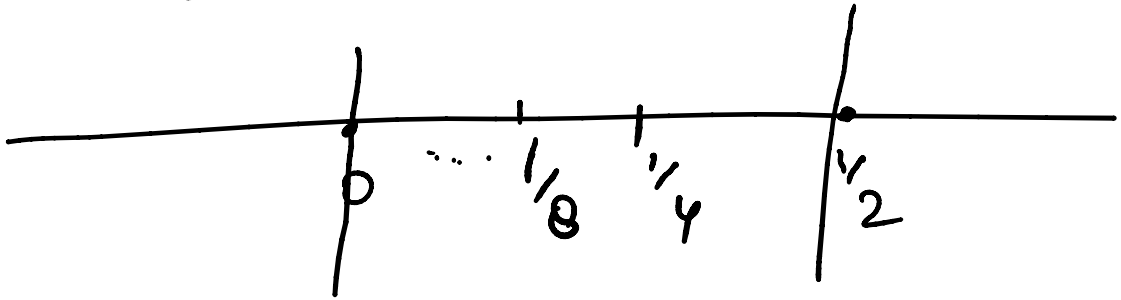
$\{2x\}$

→ supremum

→ infimum

$$\left\{ \sin \frac{n\pi}{2} \right\} = \{ 1, -1, 1, -1 \ldots \}$$

$$\left\{ \frac{1}{2^n} \right\} = \left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \ldots \right\}$$



Monotonic sequences :

1) <u>non-decreasing</u> or <u>increasing</u> $\{ a_n \}$

$$a_1 \leq a_2 \leq a_3 \leq a_4 \ldots$$

2) non-increasing or decreasing

$$a_1 \geq a_2 \geq a_3 \geq a_4 \ldots$$

$$a_1 \geq a_2 \geq a_3 \geq a_4 \cdots\cdots$$

3) Strictly increasing

$$a_1 < a_2 < a_3 < a_4 \cdots\cdots$$

4) Strictly decreasing

$$a_1 > a_2 > a_3 > a_4 \cdots\cdots$$

④ eg. $\left\{\frac{1}{n}\right\}$ $\quad 1 > \frac{1}{2} > \frac{1}{3} \cdots\cdots$

③ $\left\{2^n\right\}$ $\quad 2 < 4 < 8 \cdots\cdots$

$(-1)^{n^2}$

② $\left\{ 1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3} \ldots \right\}$

$$a_n = \begin{cases} \frac{2}{n+1} & n \text{ odd} \\ 2/n & n \text{ even} \end{cases}$$

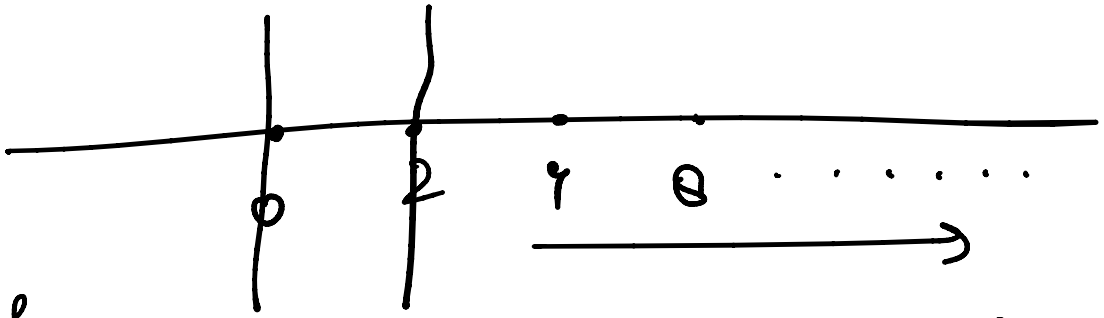
① $$a_n = \begin{cases} n & n \text{ odd} \\ n+1 & n \text{ even} \end{cases}$$

$$= \{ 1, 3, 3, 5, 5 \ldots \}$$

& $\{ [\sqrt{n}] \} = \{ 1, 1, 1, 2, 2, 2, 2, 2, 3 \ldots \}$

$\Rightarrow \{ 2^n \} = \{ 2, 4, 8 \ldots \}$

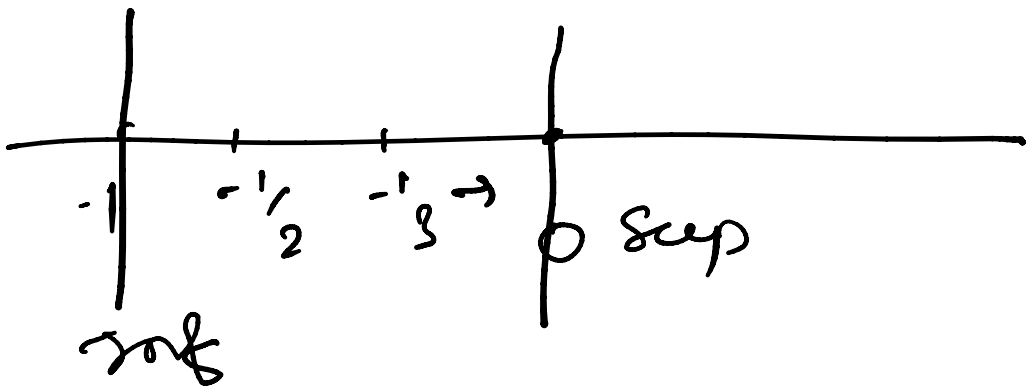
$\Rightarrow \{2^n\} = \{2, 4, 8 \ldots\}$



$\overline{R} = R \cup \{-\infty, \infty\}$

inf = 2 sup = not defined $= \infty$

$\left\{-\frac{1}{n}\right\} = \left\{-1, -\frac{1}{2}, -\frac{1}{3} \ldots\right\}$



① If $\uparrow$ $\inf\{a_n\} = a_1$ & sup = $\lim a_n$

② If $\downarrow$ $\sup\{a_n\} = a_1$ & $\inf = \lim a_n$

$$\left\{1 - \frac{1}{n}\right\} = \left\{0, \frac{1}{2}, \frac{2}{3} \ldots\right\}$$

$\inf = 0$ $\sup = 1$